

What Powers Artistly Gallery — A Plain-English Guide

The Big Picture

Artistly Gallery is a desktop application built entirely from free, open-source tools. Everything runs locally on your machine — there is no account to create, no subscription, and no data leaving your computer except during the cloud sync operations that connect to your existing Artistly account.

The short version: Python runs the whole show. FastAPI and Uvicorn handle the internal wiring. pywebview gives it a proper desktop window. SQLite stores your library data in a single local file. React and Tailwind CSS build the interface you see on screen. Playwright handles the cloud automation. Each piece has one clear job.

Python — The Foundation

Python is the programming language the entire application is written in. When you launch Artistly Gallery, Python is what actually starts up — it brings every other component to life, manages the flow of data between them, and handles everything that happens in response to your actions.

Python was the right choice for this app because it runs identically on Windows and Mac, has ready-made tools for everything the app needs to do, and can be packaged into a self-contained installer that requires nothing extra from you.

FastAPI and Uvicorn — The Internal Wiring

These two components work as a team and you can think of them together: they form the internal communication layer that sits between the interface you see and the data stored on your computer.

When you click "Sync," flag an image, change a folder, or mark a favorite, something has to receive that action, do the actual work (update the database, launch a task, read a file), and send the result back to the screen. That's what FastAPI does — it's the coordinator.

Uvicorn is what runs FastAPI in the background. It starts silently when the app launches and keeps listening for activity. It operates on an address that only exists on your own machine — nothing about it is reachable from the internet or your home network.

You never interact with either of these directly. They're the plumbing.

Playwright — Cloud Operations

Playwright is the browser automation tool that handles cloud sync tasks, bulk moves, and bulk deletes on your behalf — so you don't have to manually click through hundreds of pages on Artistly's website.

For the full explanation of how Playwright works, which browser it uses, and how login is handled, see the separate [Playwright_Explainer.pdf](#) documentation.

pywebview — The Desktop Window

pywebview is what makes Artistly Gallery feel like a proper desktop application rather than a website you open in a browser tab.

When the app launches, pywebview creates a native window on your desktop with its own icon in the taskbar, its own title bar, and normal minimize and close controls. Inside that window it displays the app's interface. From your perspective it looks and behaves like any other installed app. The fact that the interface is built with web technologies is completely invisible to you.

pywebview also provides the native file and folder picker dialogs — when you click Browse or choose a save location, the dialog that appears is your operating system's standard one, not something web-based.

SQLite — Your Local Library

SQLite is the database that stores everything the app knows about your gallery: your designs, their metadata, folder assignments, download status, favorites, sync history, and which images are flagged for upcoming actions.

It lives as a single file on your computer inside your Artistly Gallery data folder. There is no database server to install or manage — it's just a file. It never sends data anywhere and never requires an internet connection.

Because it's a single file, it's easy to back up. The Gallery Backup feature in the Control Center copies this file along with your images folder.

React 18 — The Visual Interface

React is the framework that builds and manages everything you see on screen — the gallery grid, the filter panel, the Control Center cards, the zoom modal, the theme switcher, all of it. It keeps the display in sync with your data and handles interactions like filtering, searching, and opening images smoothly without requiring full page reloads.

In most applications, React requires a separate installation process and a set of developer tools running on your machine. Artistly Gallery skips all of that. The entire interface lives inside a single file (`index.html`), and React loads automatically from the web when the app starts. There is nothing to install, nothing to compile, and no developer tools required.

Tailwind CSS — The Visual Design System

Tailwind CSS controls how everything looks — the spacing, colors, fonts, layout, and visual design of every button, card, panel, and label. It's what keeps the interface consistent and polished, and it's what powers the nine built-in themes.

Like React, Tailwind loads automatically from the web at startup rather than requiring a separate installation. The nine themes — Dark, Light, Deep Space, Medical, Inkwell, Newsprint, Carbon, Birch, and Artistly Purple — all use Tailwind and switch instantly without any page reload.